

Symbol Grounding in Neuro-Symbolic AI: A Gentle Introduction to Reasoning Shortcuts

DISI, Ph.D. course on NeSy AI

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Other conspirators of Reasoning Shortcuts



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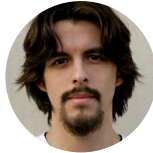
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SYMBOL GROUNDING IN NEURO-SYMBOLIC AI: A GENTLE INTRODUCTION TO REASONING SHORTCUTS

A PREPRINT

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Preprint: <https://arxiv.org/abs/2510.14538>

Outline

- 1. Symbol Grounding**
- 2. Reasoning Shortcuts**
- 3. Mitigations**
- 4. Awareness**
- 5. Resources**
- 6. Open Problems**

Symbol Grounding

[!] Two problems in NeSy AI

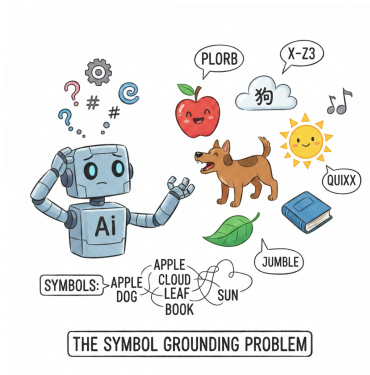
- Bridging the gap between continuous and discrete computation
- Symbol grounding problem (or latent symbol identification)

—

[!] Emile van Krieken, *"Optimisation in Neurosymbolic Learning Systems"*, Ph.D. thesis (2024).

Symbol Grounding Problem [1]

- Learning *symbols/concepts* in neural nets from data is an open problem.
- How symbols are used in neural nets is "implicit".
- NeSy AI provides logical rules upon which learned symbols are used, making them "explicit".
- Harnad anticipated this is a problem of **identification**.

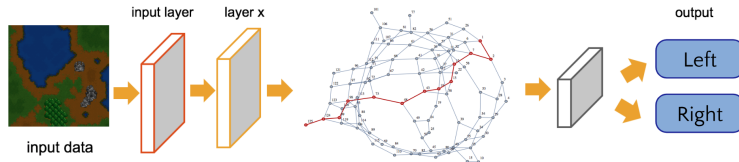


Courtesy of Gemini 3.0

[1] Harnad, "The Symbol Grounding Problem", Physica D (1990).

Use cases of NeSy AI

Combining discrete optimization models (e.g. path-finding, planning) and neural networks [2]



Enforcing logical constraints in neural networks to increase **trustworthiness** [3]



ROAD-R is an extension of the ROAD dataset with a set of 243 manually annotated requirements over the 41 labels grouped into agents, actions and locations. The requirements are logical constraints provided in disjunctive normal form and express background knowledge applicable in autonomous driving scenarios, such as:

- A traffic light cannot be red and green at the same time.
- A vehicle lane cannot be a parking lot,
- A traffic light cannot move,
- If an agent is crossing, it is either a pedestrian or a cyclist.

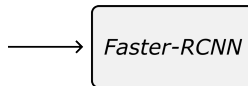


[2] Niepert et al., "Implicit MLE: Backpropagating Through Discrete Exponential Family Distributions", NeurIPS (2021)

[3] Giunchiglia et al., "ROAD-R: The Autonomous Driving Dataset with Logical Requirements", MLJ (2024)

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e.g., autonomous driving with **safety constraints**

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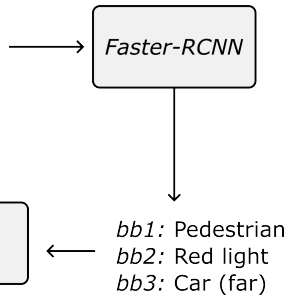


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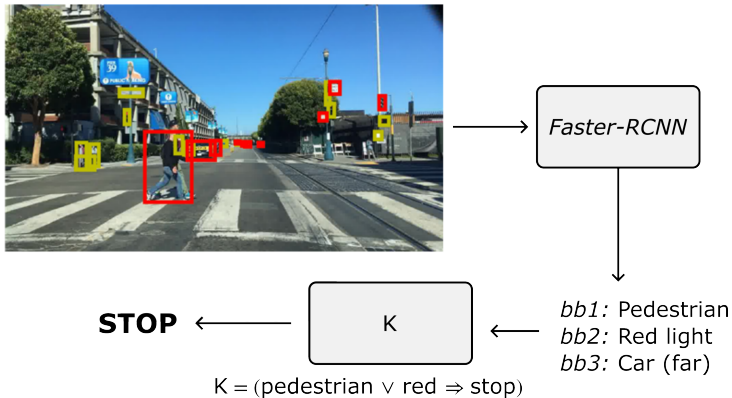
bb1: Pedestrian
bb2: Red light
bb3: Car (far)

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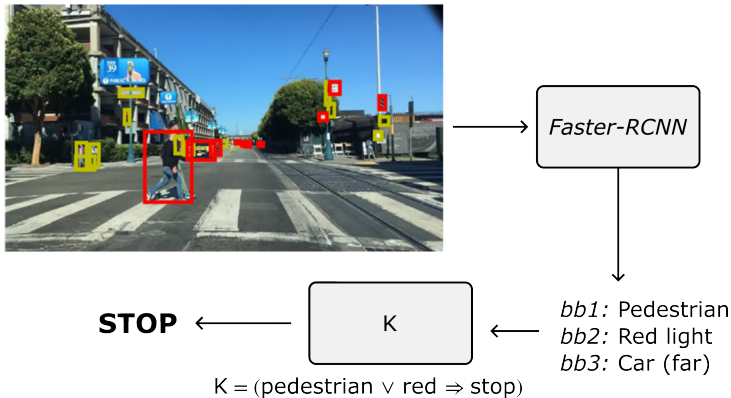


$K = (\text{pedestrian} \vee \text{red} \Rightarrow \text{stop})$

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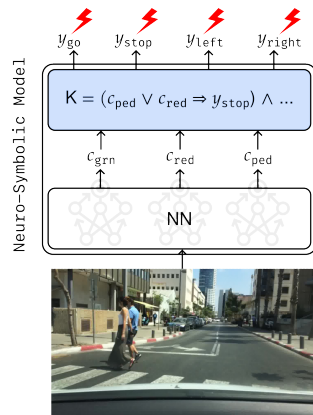


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Examples: DeepProbLog, Semantic Loss, and Logic Tensor Networks

Let's formalize it!

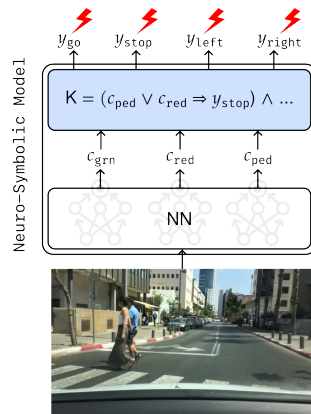


⚡: supervised

Let's formalize it!

1. Neural extraction of concepts:

$$\mathbf{f}_\theta : \mathbb{R}^n \rightarrow \{0, 1\}^k, \quad \mathbf{f}_\theta : \mathbf{x} \mapsto \mathbf{c} \quad (1)$$



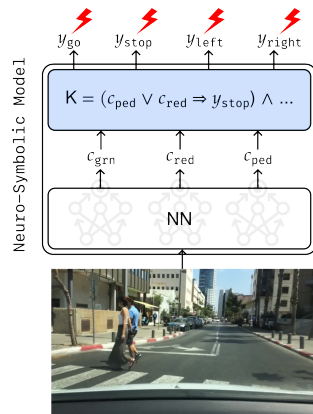
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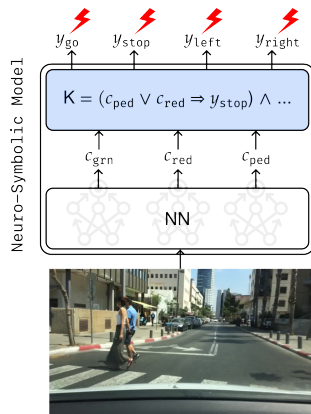
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NeSy model = combine neural processing for concept extraction with symbolic rules:

$$p_\theta(\mathbf{y} \mid \mathbf{x}; K) := (\beta_K \circ \mathbf{f}_\theta)(\mathbf{x}) \quad (3)$$



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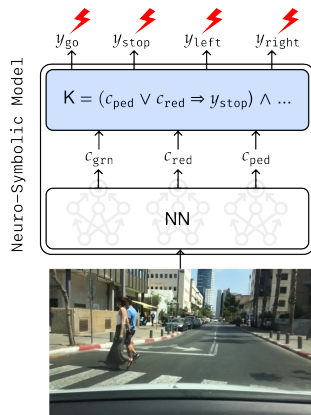
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■ Trained to achieve maximum log-likelihood:

$$\max_{\theta} \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}} \log p_\theta(\mathbf{y} \mid \mathbf{x}; K) \quad (4)$$



⚡: supervised

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*"if my NeSy model predicts correct actions in all examples of autonomous driving, **then the concepts are good!** For instance, `red_light` = $\top$ **if and only if** there is a red traffic light in the dashcam image!"*

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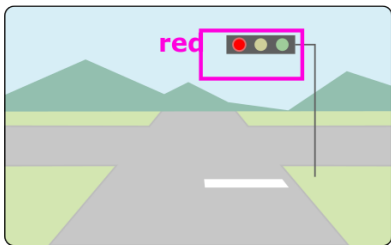
- This would be ideal: concepts with the right semantics **generalize** to new tasks (as required by, e.g., NeSy verification [4]) and support **interpretability**.

[4] Xie et al., "Neuro-Symbolic Verification of Deep Neural Networks", IJCAI (2022).

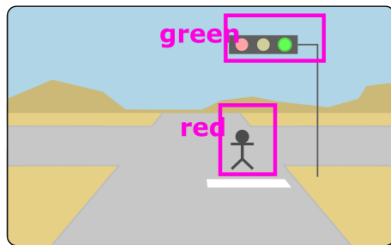


Reasoning Shortcuts

$$K_1 = (\text{pedestrian} \vee \text{red} \Rightarrow \text{stop})$$



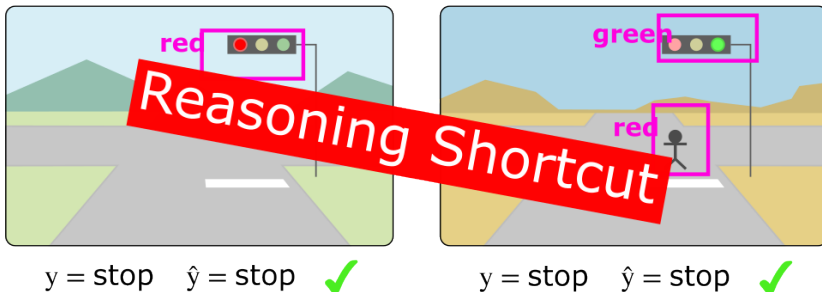
$y = \text{stop}$ $\hat{y} = \text{stop}$ ✓



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■ Task: predict stop vs. go using concepts "pedestrian", "red", and "green".

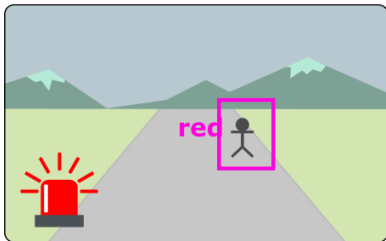
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■ Task: predict stop vs. go using concepts “pedestrian”, “red”, and “green”.

Reasoning Shortcut: Perfect accuracy by predicting pedestrians as red lights!!!

$$K_2 = (\text{emergency} \wedge \neg \text{pedestrian} \Rightarrow \text{go}) \wedge K_1$$



$\hat{y} = \text{go}$ **X**



■ Task: ...but now if there is an emergency, we can ignore “red” lights

Theoretical Description of Reasoning Shortcuts

Two pictures:

- Identifiability Perspective (today) [5,6]
- Statistical Learning Theory Perspective [7,8]

— — — — —

[5] Marconato et al., *"Not All Neuro-Symbolic Concepts Are Created Equal: Analysis and Mitigation of Reasoning Shortcuts"*, NeurIPS (2023)

[6] Bortolotti et al., *"Shortcuts and Identifiability in Concept-based Models from a Neuro-Symbolic Lens"*, NeurIPS (2025).

[7] Yang et al., *"Analysis for Abductive Learning and Neural-Symbolic Reasoning Shortcuts"*, ICML (2024).

[8] He et al., *"A Learnability Analysis on Neuro-Symbolic Learning"*, NeurIPS (2025).

Overview of the Identifiability Picture

- ① RSs arise because of non-identifiability of ground-truth concepts
- ②
- ③ We can count how many deterministic RSs are present (shared)
- ④ Mitigations act on the count, reducing # of deterministic RSs

Symbol Grounding
○○○○○○○

Reasoning Shortcuts
○○○○●○○○

Mitigations
○○○○○○○

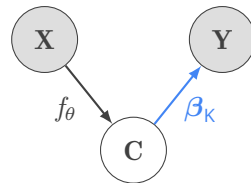
Awareness
○○○○○

Resources
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Open Problems
○○○○○○○

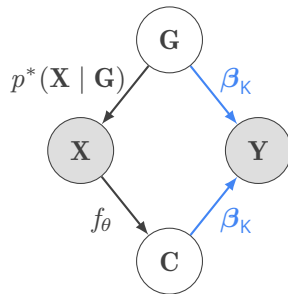
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- NeSy model map inputs X to *discrete concepts* C (via NN_θ) and use knowledge (via β_K) to map to labels Y



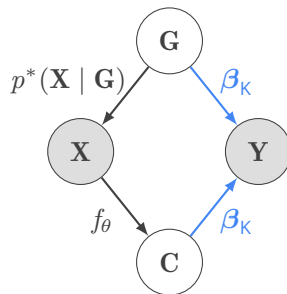
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- Ground-truth *discrete concepts* $\mathbf{G}$ cause inputs $\mathbf{X}$ (via $p^*(\mathbf{X} | \mathbf{G})$) and labels $\mathbf{Y}$ (via β_K)



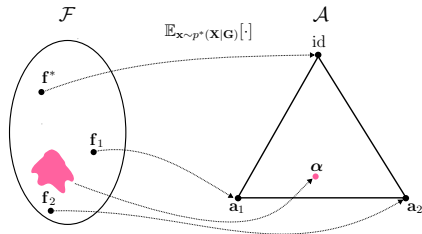
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 - Ground-truth *discrete concepts* $\mathbf{G}$ cause inputs $\mathbf{X}$ (via $p^*(\mathbf{X} | \mathbf{G})$) and labels $\mathbf{Y}$ (via β_K)
 - (RSs) NeSy models attaining *optimal log-likelihood* may not learn correct concepts
- ≡ **non-identifiability** of learned concepts: $\mathbf{C} \neq \mathbf{G}$



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Concept Remapping Distributions



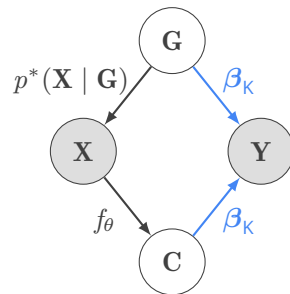
- We introduce *concept remapping distributions*

$$\alpha(\mathbf{G}) := p_{\theta}(\mathbf{C} \mid \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p^*(\mathbf{X}|\mathbf{G})}[\mathbf{f}_{\theta}(\mathbf{x})] \quad (5)$$

- Simplex structure of $\mathcal{A}$:

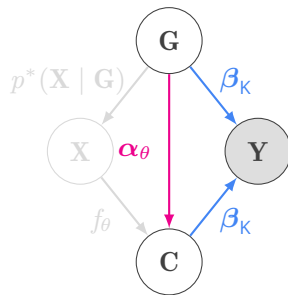
$$\forall \alpha \in \mathcal{A}, \quad \alpha(\mathbf{g}) = \sum_{\mathbf{a}_i \in \text{Vert}(\mathcal{A})} \lambda_i \mathbf{a}_i(\mathbf{g}) \quad (6)$$

Reasoning Shortcuts = non-identifiability of latent concepts



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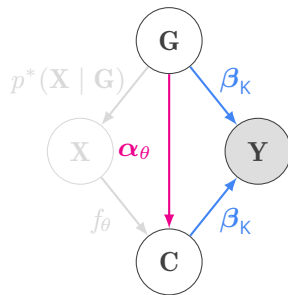
- We leverage maps $\alpha: \mathbf{G} \mapsto \mathbf{C}$ to describe the problem:



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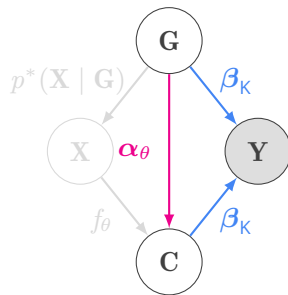
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■ We leverage maps $\alpha: \mathbf{G} \mapsto \mathbf{C}$ to describe the problem:

- $\alpha = \text{id}$, intended solution
- $\alpha \neq \text{id}$ and optimal, Reasoning Shortcut (RS)



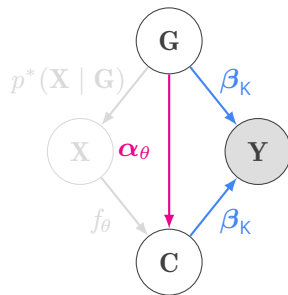
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When is α a RS?

$$(\beta_K \circ \alpha)(g) = \beta_K(g)$$



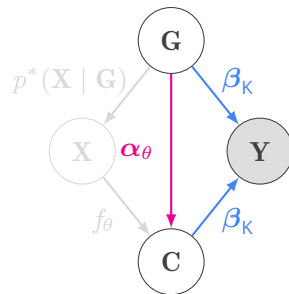
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When is α a RS?

$$\bigwedge_{\mathbf{g} \in \mathcal{D}_{\mathbf{G}}} (\beta_{\mathbf{K}} \circ \alpha)(\mathbf{g}) = \beta_{\mathbf{K}}(\mathbf{g})$$



Reasoning Shortcuts = non-identifiability of latent concepts

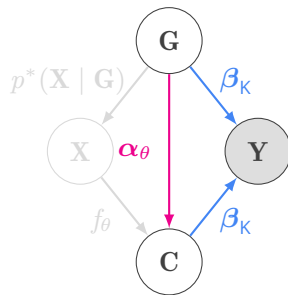
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Theorem. (Number of deterministic RSs)

The number of (deterministic) RSs is:

$$\sum_{\alpha \in \text{Vert}(\mathcal{A})} \mathbb{1}\left\{ \bigwedge_{\mathbf{g} \in \mathcal{D}_{\mathbf{G}}} (\beta_{\mathbf{K}} \circ \alpha)(\mathbf{g}) = \beta_{\mathbf{K}}(\mathbf{g}) \right\} - 1 \geq 0$$



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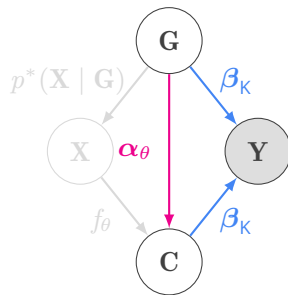
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■ Same counts holds for all NeSy models



MnistAdd: a prototypical task

$$\begin{cases} \boxed{0} + \boxed{1} = 1 \\ \boxed{0} + \boxed{2} = 2 \end{cases}$$

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Knowledge: $\beta_K(c_1, c_2) = c_1 + c_2$.

Support: $\mathcal{D}_G = \{(0, 1), (1, 0), (0, 2), (2, 0)\}$

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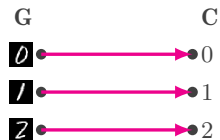
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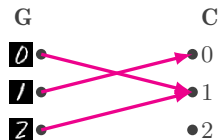


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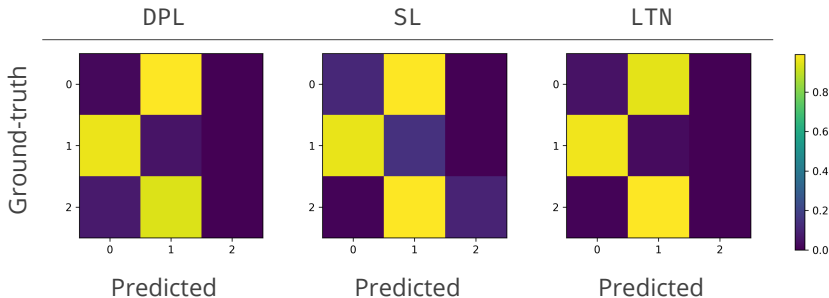
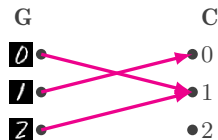


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Causes of RSs can be read off of it:

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Causes of RSs can be read off of it:

($\mathcal{K}$) Structure of **knowledge** $\mathbf{K}$, via $\beta_{\mathbf{K}}$

($\mathcal{D}$) Structure of **data set**, via $\mathcal{D}_{\mathbf{G}}$

$$\sum_{\alpha \in \text{Vert}(\mathcal{A})} \mathbb{1}\{ \bigwedge_{\mathbf{g} \in \mathcal{D}_{\mathbf{G}}} \underbrace{(\beta_{\mathbf{K}} \circ \alpha)(\mathbf{g}) = \beta_{\mathbf{K}}(\mathbf{g})}_{\text{objective}} \} - 1 \geq 0$$

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Ⓚ Structure of **knowledge** $\mathbf{K}$, via $\beta_{\mathbf{K}}$

ⓓ Structure of **data set**, via $\mathcal{D}_{\mathbf{G}}$

ℒ **Loss**, via optimality of *objective*

$$\sum_{\alpha \in \text{Vert}(\mathcal{A})} \mathbb{1}\{ \bigwedge_{\mathbf{g} \in \mathcal{D}_{\mathbf{G}}} \underbrace{(\beta_{\mathbf{K}} \circ \alpha)(\mathbf{g}) = \beta_{\mathbf{K}}(\mathbf{g})}_{\text{objective}} \} - 1 \geq 0$$

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Ⓐ **Architecture bias** of the model, via α

Mitigations

Handling RSs

Table 2. **Overview of mitigation strategies.** The columns indicate, for each strategy, what component(s) of Eq. (44) it targets (**Target**), whether it requires extra annotations (**Annot**) and guarantees reducing the count in Eq. (24) (**Count**), and how well it worked on different NeSy Tasks. Specifically, “✓” means that it prevents all RSs, “✗” that it fails to do so, and “?” that it has not been evaluated. **Note:** With MNIST-Add * we refer to the family of datasets based on the MNIST-Add benchmark (Manhaeve et al., 2018).

Strategy	Properties			Arithmetic	Logic		High-Stakes
	Target	Annot	Count	MNIST-Add *	Kandinsky	Clevr	BDD-OIA
Concept Supervision (Koh et al., 2020)	ℒ	✓	✓	✓	✓	✓	✗
Multi-task Learning (Caruana, 1997)	K	✓	✓	✓	?	?	?
Prototype-based grounding (Snell et al., 2017)	ℒ	✓	✓	✓	✓	✓	✗
Weak Supervision (Yang et al., 2024)	ℒ	✗	✗	✓	?	?	✗
Entropy Maximization (Manhaeve et al., 2021a)	ℒ	✗	✗	?	?	?	✗
Smoothing (Szegedy et al., 2016)	ℒ	✗	✗	?	?	?	?
Reconstruction (Kingma, 2013)	ℒ	✗	✓	✓	?	?	?
Contrastive Learning (Chen et al., 2020)	ℒ	✗	✗	?	?	?	?
Disentanglement (Suter et al., 2019)	ℱ	✗	✓	?	✗	?	?

■ Prototype-based grounding recently proposed by Andolfi and Giunchiglia [9]

[9] Andolfi and Giunchiglia, “*Right for the Right Reasons: Avoiding Reasoning Shortcuts via Prototypical Neurosymbolic AI*”, NeurIPS (2025).

Table 1: Impact of different mitigation strategies on the number of deterministic optima: R is reconstruction, C supervision on C, MTL multi-task learning, and DIS disentanglement. All strategies reduce the number of α 's in [Eq. \(6\)](#), sometimes substantially, but require different amounts of effort to be put in place. Actual counts for our data sets are reported in [Appendix C.2](#).

MITIGATION	REQUIRES	CONSTRAINT ON α	ASSUMPTIONS	RESULT
None	–	$\bigwedge_{\mathbf{g} \in \text{supp}(\mathbf{G})} ((\beta_K \circ \alpha)(\mathbf{g}) = \beta_K(\mathbf{g}))$	A1, A2	Theorem 2
MTL	Tasks	$\bigwedge_{\mathbf{g} \in \text{supp}(\mathbf{G})} \bigwedge_{t \in [T]} ((\beta_{K(t)} \circ \alpha)(\mathbf{g}) = \beta_{K(t)}(\mathbf{g}))$	A1, A2	Proposition 4
C	Sup. on C	$\bigwedge_{\mathbf{g} \in \mathcal{S} \subseteq \text{supp}(\mathbf{G})} \bigwedge_{i \in I} (\alpha_i(\mathbf{g}) = g_i)$	A1	Proposition 5
R	–	$\bigwedge_{\mathbf{g}, \mathbf{g}' \in \text{supp}(\mathbf{G}) : \mathbf{g} \neq \mathbf{g}'} (\alpha(\mathbf{g}) \neq \alpha(\mathbf{g}'))$	A1, A3	Proposition 6

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■ Other unsupervised (*Reconstruction, Disentanglement, Entropy*) can help, but not enough!

Multi-tasking in MnistAdd

$$\left\{ \begin{array}{l} \boxed{0} + \boxed{1} = 1 \\ \boxed{0} + \boxed{2} = 2 \\ \boxed{0} \times \boxed{1} = 0 \\ \boxed{0} \times \boxed{2} = 0 \end{array} \right.$$

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Knowledge: $\beta_1(c_1, c_2) = c_1 + c_2$, $\beta_2(c_1, c_2) = c_1 \times c_2$

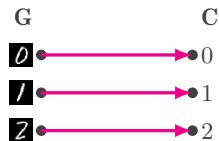
G
 $\boxed{0} \bullet$
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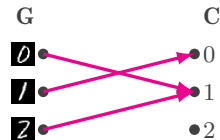
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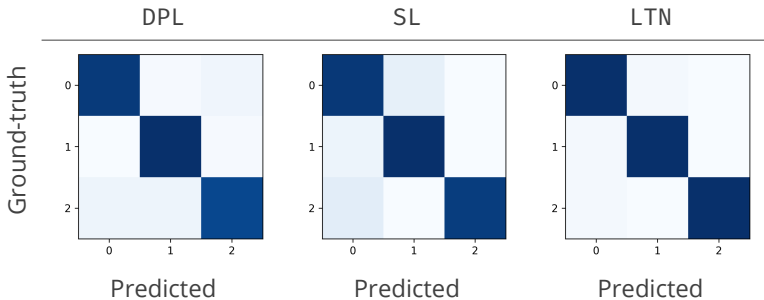
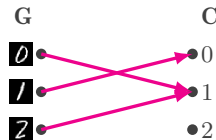
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Q: Do you avoid all RSs by zeroing deterministic RSs?

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Short answer: No! It depends on the symbolic layer β_K :

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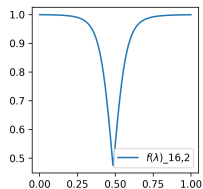
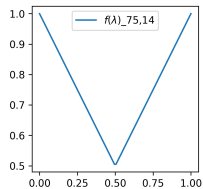
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Extremality condition

The inference layer β_K satisfies extremality if, for all $\lambda \in (0, 1)$ and for all $\mathbf{c} \neq \mathbf{c}' \in \mathcal{C}$ such that $\operatorname{argmax}_{y \in \mathcal{Y}} \beta_K(\mathbf{c})_y \neq \operatorname{argmax}_{y \in \mathcal{Y}} \beta_K(\mathbf{c}')_y$, where $\beta_K(\cdot)_y$ is the y -th component of β_K , we have

$$\max_{y \in \mathcal{Y}} \beta_K(\lambda \mathbb{1}\{\mathbf{C} = \mathbf{c}\} + (1 - \lambda) \mathbb{1}\{\mathbf{C} = \mathbf{c}'\})_y < \quad (7)$$
$$\max \left(\max_{y \in \mathcal{Y}} \beta_K(\mathbf{c})_y, \max_{y \in \mathcal{Y}} \beta_K(\mathbf{c}')_y \right).$$



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① Empirical Label Risk $\mathcal{R}_Y$ vs Empirical Concept Risk $\mathcal{R}_C$

$$\mathcal{R}_Y = -\frac{1}{|\mathcal{D}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}} [\log p_{\theta}(\mathbf{y} \mid \mathbf{x}; \mathbf{K})], \quad \mathcal{R}_C = -\frac{1}{|\mathcal{D}|} \sum_{(\mathbf{x}, \mathbf{g}) \in \mathcal{D}} [\log p_{\theta}(\mathbf{C} = \mathbf{g} \mid \mathbf{x})] \quad (8)$$

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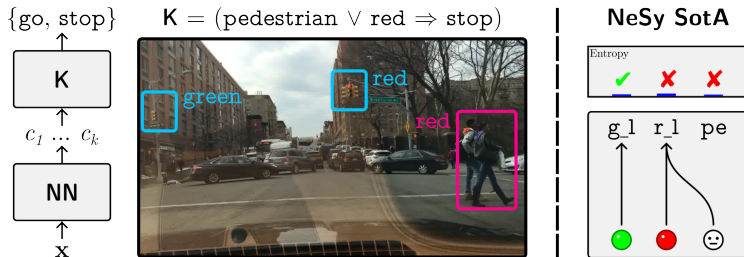
② The **Empirical Knowledge Complexity** (EKC, the higher the better) indicates if, in the limit of infinite data,

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③ When EKC is maximum, one can provide bounds on (1) **ERM of $\mathcal{R}_C$** , also (2) **for different NeSy modules** (like ABL), and (3) **for different mitigations** (e.g., label smoothing)

Awareness

■ Some RSs can be **over-confident** on concepts



[★] “*bears Make Neuro-Symbolic Models Aware of their Reasoning Shortcuts*”, E. Marconato, S. Bortolotti, E. van Krieken, A. Vergari, A. Passerini, and S. Teso at UAI (2024)

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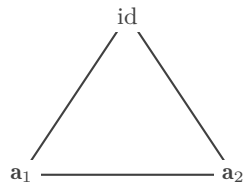
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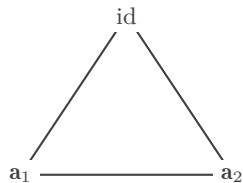
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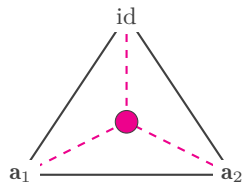
$$\mathbf{a}_1, \mathbf{a}_2, \text{id} \in \text{Vert}(\mathcal{A}) \quad (10)$$



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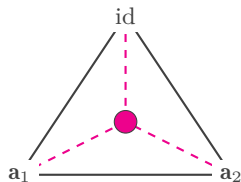
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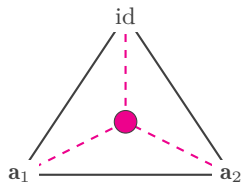


- Example on MNIST digit recognition ($\mathbf{a} : \{0, 1, \dots, 9\} \rightarrow \{0, 1, \dots, 9\}$)

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$$\alpha = 0.5\mathbf{a}_1 + 0.5\mathbf{a}_2 \implies \alpha(\mathbf{0}) = \begin{pmatrix} 0.5 \\ 0.5 \\ 0.0 \end{pmatrix}, \alpha(\mathbf{1}) = \begin{pmatrix} 0.0 \\ 1.0 \\ 0.0 \end{pmatrix} \quad (13)$$

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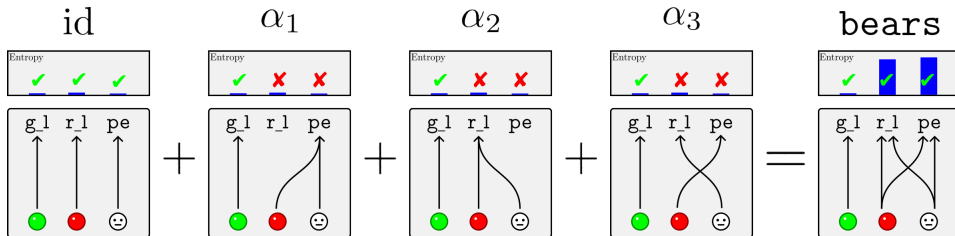
■ bears = *Deep Ensembles* + *diversification*

$$\max_{\theta_t} \mathcal{L}(\mathbf{x}, \mathbf{y}; K, \theta_t) + \gamma_1 \cdot \text{KL}(\mathbf{f}_{\theta_t}(\mathbf{x}) \parallel \frac{1}{t} \sum_{j=1}^t \mathbf{f}_{\theta_j}(\mathbf{x})) + \gamma_2 \cdot H(\mathbf{f}_{\theta_t}(\mathbf{x})) \quad (14)$$

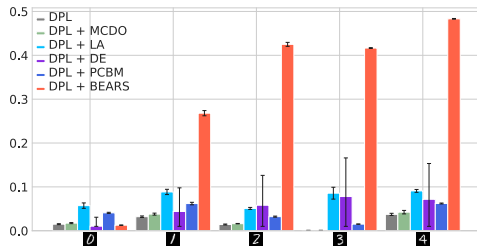
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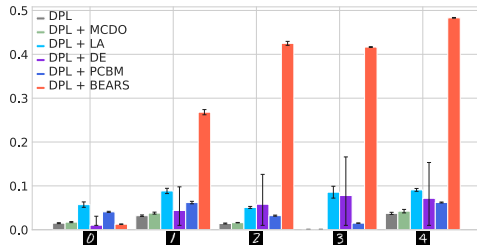


bears improves awareness and lessens the cost of mitigations!

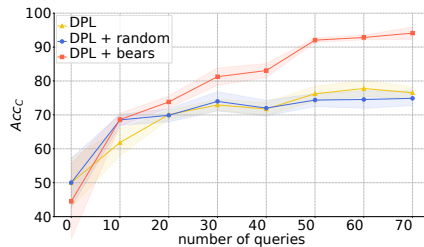


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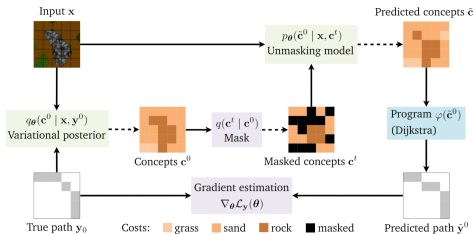
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bears allows to select relevant annotation faster!

NeSy Diffusion Models [12] for awareness

Loss spreads concept predictions, thus capturing non-deterministic α 's

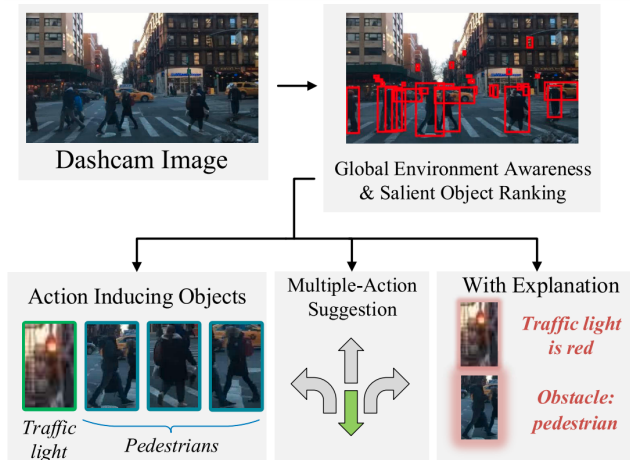


$$\mathcal{L}_{\text{NESYDM}} = \mathbb{E}_{t \sim [0,1], q_{\theta}(\mathbf{c}^0 | \mathbf{x}, \mathbf{y}^0), q(\mathbf{c}^t | \mathbf{c}^0)} \left[\underbrace{\frac{\alpha'_t}{1 - \alpha_t} \sum_{i=1}^C \log p_{\theta}(\tilde{c}_i^0 = c_i^0 | \mathbf{c}^t, \mathbf{x})}_{\mathcal{L}_{\mathbf{c}}: \text{concept unmasking loss}} + \underbrace{\alpha'_t \sum_{i=1}^Y \log \sum_{\tilde{\mathbf{c}}^0} p_{\theta}(\tilde{\mathbf{c}}^0 | \mathbf{c}^t, \mathbf{x}) \mathbb{1}[\varphi(\tilde{\mathbf{c}}^0)_i = y_i^0]}_{\mathcal{L}_{\mathbf{y}}: \text{output unmasking loss}} \right] - \underbrace{H[q_{\theta}(\mathbf{c}^0 | \mathbf{y}^0, \mathbf{x})]}_{\mathcal{L}_{H[q]}: \text{variational entropy}}$$

[12] van Krieken, E., Minervini, P., Ponti, E., & Vergari, A. "Neurosymbolic Diffusion Models". NeurIPS (2025).

Resources

BDD-OIA: Reasoning Shortcuts in autonomous driving [10]



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■ Predict one or more actions:

- move_forward / stop
- turn_left
- turn_right

■ 21 concepts including:

- red_light / green_light
- obstacle / road_clear
- ...

[10] Xu et al., "Explainable Object-induced Action Decision for Autonomous Vehicles", CVPR (2020)

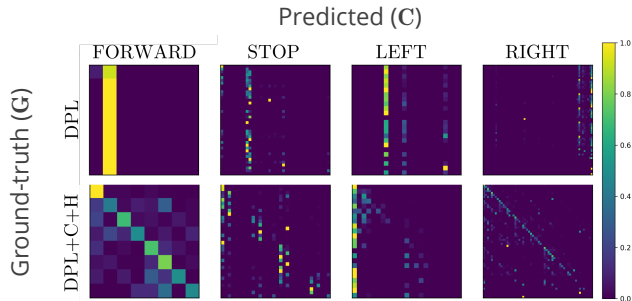
BDD-OIA: Reasoning Shortcuts in autonomous driving [10]

■ Predict one or more actions:

- move_forward / stop
- turn_left
- turn_right

■ 21 concepts including:

- red_light / green_light
- obstacle / road_clear
- ...



Top: No supervision on concepts

Bottom: Full supervision on concepts + entropy

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rsbench: <https://unitn-sml.github.io/rsbench>

	Task	Data			Properties		
		Gen	OOD	ConL	Cplx x	Cplx K	Amb K
Arithmetic	MNMath (new)	✓	✓	✓	✗	✓	✗
	MNAdd-Half	✗	✓✓	✗	✗	✗	–
	MNAdd-EvenOdd	✗	✓✓	✓✓	✗	✗	–
Logic	MNLogic (new)	✓	✓	✓	✗	✓	✗
	Kand-Logic	✓	✓	✓	✗	✓	✓
	CLE4EVR	✓	✓	✓✓	✓	✗	✓
High Stakes	BDD-OIA	✗	✗	✗	✓	✓	✓
	SDD-OIA (new)	✓	✓✓	✓	✓	✓	✓

[★] Bortolotti et al., “rsbench: A Neuro-Symbolic Benchmark Suite for Concept Quality and Reasoning Shortcuts”, NeurIPS (2024)

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TASK	TRAINING SET	KNOWLEDGE K	EXAMPLE RS	IMPACT
MNLogic	 $\oplus$  $\oplus$  $\oplus$  = 0	Formula must hold.	$\begin{cases} \square \rightarrow 1 \\ \blacksquare \rightarrow 0 \end{cases}$	 $\wedge$  = 1
Kand-Logic	  = 1	Pattern must hold.	$\begin{cases} \square \rightarrow \text{red} \\ \triangle \rightarrow \text{yel} \\ \circ \rightarrow \text{blu} \end{cases}$	  = 0
CLE4EVR	 = 0,  = 1	Same color and shape?	$\begin{cases} \blacksquare \rightarrow \blacksquare \\ \square \rightarrow \square \\ \circ \rightarrow \blacksquare \end{cases}$	 = 1

High Stakes	BDD-OIA	✗	✗	✗	✓	✓	✓
	SDD-OIA (new)	✓	✓✓	✓	✓	✓	✓

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TASK	TRAINING SET	KNOWLEDGE K	EXAMPLE RS	IMPACT			
SDD-OIA	 = stop	Traffic laws.	$\left\{ \begin{array}{l} \text{pedestrian} \rightarrow \\ \text{red_light} \\ \text{green_light} \rightarrow \\ \text{green_light} \end{array} \right.$	 = go			
BDD-OIA	 = stop	Traffic laws.	$\left\{ \begin{array}{l} \text{pedestrian} \rightarrow \\ \text{red_light} \\ \text{green_light} \rightarrow \\ \text{green_light} \end{array} \right.$	 = go			
High Stakes	BDD-OIA	✗	✗	✗	✓	✓	✓
	SDD-OIA (new)	✓	✓✓	✓	✓	✓	✓

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All models learn poor concepts due to Reasoning Shortcuts!

Results on SDD-OIA

	$mF_1(Y)(\uparrow)$	$mF_1(C)(\uparrow)$	$ood\text{-}mF_1(Y)(\uparrow)$
DPL	0.80 ± 0.01	0.49 ± 0.03	0.62 ± 0.09 (−0.18)
LTN	0.82 ± 0.04	0.46 ± 0.04	0.72 ± 0.06 (−0.10)
NN*	0.93 ± 0.18	0.44 ± 0.02	0.47 ± 0.19 (−0.46)
CLIP*	0.90 ± 0.09	0.43 ± 0.04	0.81 ± 0.06 (−0.09)

■ NeSy models ■ Neural nets

* Linear probing with TCAV

Open Problems

Extensions & Potential Future Work

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1. RSs affect NeSy predictors ($x \mapsto c \mapsto y$), but **may exist in other NeSy architectures** [11]
[11] Wang, P. W., Donti, P., Wilder, B., & Kolter, Z. "Satnet: Bridging deep learning and logical reasoning using a differentiable satisfiability solver", ICML (2019).

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4. **RSs may appear also in LLMs as symbols hallucinations**, though evidence is scattered and RSs have not been characterized yet [13]
[13] Stein, A., Naik, A., Velingker, N., Naik, M., & Wong, E. "Neuro-symbolic programming in the age of foundation models: Pitfalls and opportunities", PMLR (2025).

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Correct symbol grounding has more symmetries than original version in NeSy predictors.

Intended Semantics (IS), denoted as $(\alpha_{IS}, \beta_{IS}) \sim (\text{id}, \beta_K)$, if:

$$\alpha_{IS}(g) = (\psi \circ P_\pi \circ \text{id})(g), \quad \psi(c) = (\psi_1(c_1), \dots, \psi_k(c_k)), \quad P_\pi P_{\pi-1} = I \quad (15)$$

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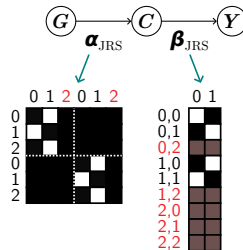
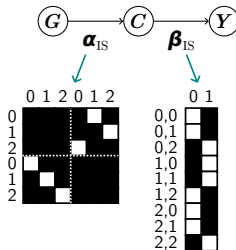
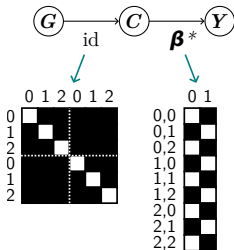


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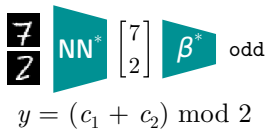


Diagram illustrating a concept-based model where both the neural network and the concept are known (or supervised). The input is a vector $\begin{bmatrix} 7 \\ 2 \end{bmatrix}$, which is processed by a known neural network NN^* to produce a known concept β^* . The output is $y = (c_1 + c_2) \bmod 2$, where c_1 and c_2 are the components of the concept vector.

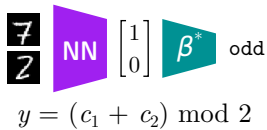


Diagram illustrating a concept-based model where the neural network is learned (purple) and the concept is known (teal). The input is a vector $\begin{bmatrix} 7 \\ 2 \end{bmatrix}$, which is processed by a learned neural network NN to produce a known concept β^* . The output is $y = (c_1 + c_2) \bmod 2$.

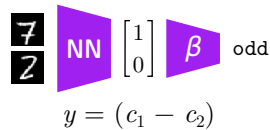


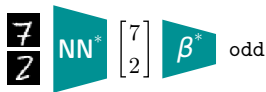
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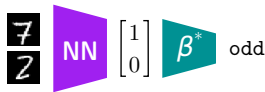
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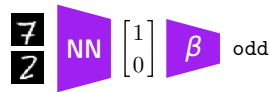
The diagram shows a black square containing the vector $\begin{bmatrix} 7 \\ 2 \end{bmatrix}$ next to a teal trapezoid labeled NN^* . This is followed by a teal trapezoid labeled β^* and the word "odd".

$$y = (c_1 + c_2) \bmod 2$$



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Theorem. (*Identifiability up to alignment*)

Under technical assumptions + extremality, it holds that:

$$(\#JRSs = 0) \implies \text{CBMs learn aligned concepts}$$

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About Neural Networks and Language Models

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👉 You can use (and if you want, also expand) `rsbench` to study controlled settings with RSs

Thank you for your attention 🔥



Review



rsbench